



IE3090

Network Programming

3rd Year, Semester I

<Assignment>

< RemoteOps: A Remote System Monitoring and Management Tool over TCP/IP >

Submitted to
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Bachelor of Science Special Honors Degree in Information Technology

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

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Personalized Configuration

The personalized filenames were used to create a unique RemoteOps project directory for registration number IT24100970. The project includes the source file of Agent, the source file of Controller, the Makefile that I modified, README, design diary, prompt log and the Agent storage directory that I modified. Git version control was set up and connected to GitHub using SSH authentication; the initial project structure and personalized configuration were successfully committed and distributed.

Registration ID	IT24100970
Agent TCP Port	9410
Agent source file	agent_970.c
Controller source file	Controller_970.c
Makefile	Makefile_970
SID	0790
SID Tag	SID:0790
Authentication Token	OPS-0790

Table 1- Unique ID

Architecture Overview

RemoteOps system follows a client-server architecture considering main **Agent & Controller** programs. Agent acts as server & runs on managed Linux machine & Controller acts as client used by administrator. TCP is the main control communication between agent & controller program, which provides reliable & ordered delivery for authentication, commands, responses, & file transfers. UDP is used for periodic system monitoring after controller sends MONITOR START.

Agent listens on unique TCP port 9410 & accepts multiple controller connections. Agent enables concurrency by a thread per client. Each controller accepted by agent executed in its own POSIX thread.

Multiple controllers can be connected & execute commands independently at same time. Each client thread handles authentication, command, file transfer, monitoring control.

Uploaded files are stored in personalized agent directory, & important tasks are recorded in a timestamped log file.

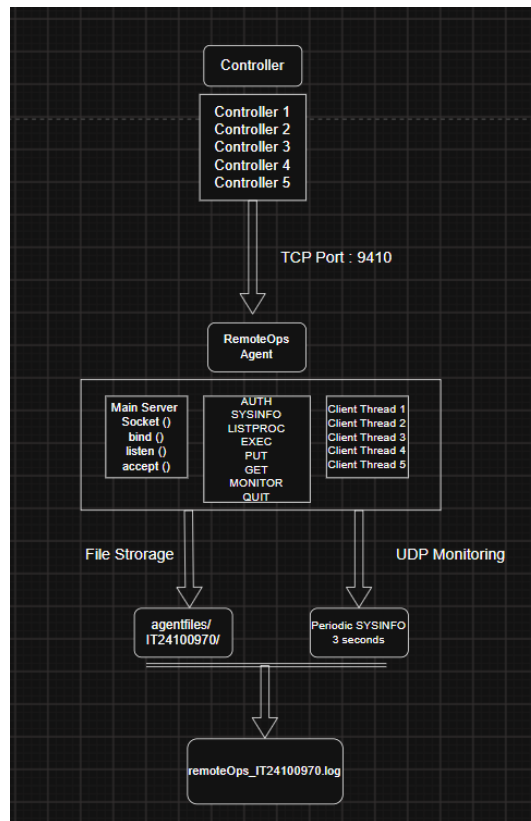


Figure 1- Client - Server Architecture

Development & Implementation

Step 1: Project Setup & File Structure

RemoteOps was created using personalized file names & values based on IT24100970. The required source files, Makefile, Readme, design diary, storage directory, & Git repository were prepared. Git was also connected to command prompt via SSH so that each development stage could be committed & tracked separately.

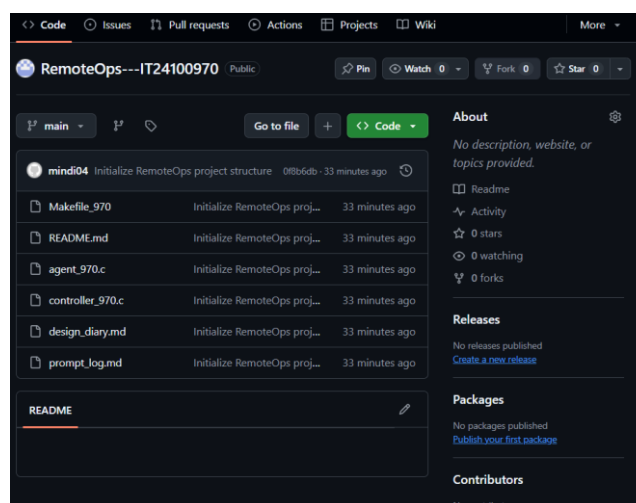


Figure 2- Git Repository

Step 2: TCP Agent Setup

Agent created as a TCP server using the BSD socket API. It creates a socket, binds to port 9410, Listens for incoming connections, & accepts a Controller connection. Listening port was verified using ss command.

```
mindio4@localhost:~/RemoteOps$ nano agent_970.c
mindio4@localhost:~/RemoteOps$ nano agent_970.c
mindio4@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindio4@localhost:~/RemoteOps$ ls -l
total 28
-rwxr-xr-x. 1 mindio4 mindio4 17376 Oct  5 15:06 agent_970
-rw-r--r--. 1 mindio4 mindio4 2103 Oct  5 14:55 agent_970.c
drwxr-xr-x. 3 mindio4 mindio4  24 Oct  5 09:59 agentfiles
-rw-r--r--. 1 mindio4 mindio4   0 Oct  5 09:57 controller_970.c
-rw-r--r--. 1 mindio4 mindio4   0 Oct  5 09:58 design_diary.md
-rw-r--r--. 1 mindio4 mindio4   0 Oct  5 09:57 Makefile_970
-rw-r--r--. 1 mindio4 mindio4   0 Oct  5 09:58 prompt_log.md
-rw-r--r--. 1 mindio4 mindio4  488 Oct  5 11:06 README.md
mindio4@localhost:~/RemoteOps$ ./agent_970
Socket created successfully.
Agent bound to TCP port 9410.
RemoteOps Agent is listening on port 9410...
Controller connected from 127.0.0.1:46586
Connection closed.
mindio4@localhost:~/RemoteOps$ |
```

Figure 3- server clarification

```
mindio4@localhost:~$ ss -tlnp | grep 9410
LISTEN 0      5            0.0.0.0:9410      0.0.0.0:*        users:(("agent_970",pid=34103,fd=3))
mindio4@localhost:~$ mkdir RemoteOps
mkdir: cannot create directory 'RemoteOps': File exists
mindio4@localhost:~$ cd RemoteOps
mindio4@localhost:~/RemoteOps$ ss -tlnp | grep 9410
LISTEN 0      5            0.0.0.0:9410      0.0.0.0:*        users:(("agent_970",pid=34103,fd=3))
mindio4@localhost:~/RemoteOps$ ss -tlnp | grep 9410
LISTEN 0      5            0.0.0.0:9410      0.0.0.0:*        users:(("agent_970",pid=34103,fd=3))
mindio4@localhost:~/RemoteOps$ nc 127.0.0.1 9410

^C
mindio4@localhost:~/RemoteOps$ |
```

Figure 4 - Port Verification

Step 3: TCP Controller Setup

Controller was implemented as TCP client. It creates a socket & connects to the Agent using IP address 127.0.0.1 & port 9410. This confirmed that the agent & controller could establish a basic TCP connection successfully.

```

mindio4@localhost:~/RemoteOps$ ./agent_970
Socket created successfully.
Agent bound to TCP port 9410.
RemoteOps Agent is listening on port 9410...
Controller connected from 127.0.0.1:46768
Connection closed.
mindio4@localhost:~/RemoteOps$ |

```

Figure 5 - Agent Listening

```

mindio4@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970
mindio4@localhost:~/RemoteOps$ ls -l controller_970
-rwxr-xr-x. 1 mindio4 mindio4 17176 Oct  5 15:56 controller_970
mindio4@localhost:~/RemoteOps$ ./controller_970
Controller socket created successfully.
Connecting to Agent at 127.0.0.1:9410...
Connected to RemoteOps Agent successfully.
Controller connection closed.
mindio4@localhost:~/RemoteOps$ |

```

Figure 6 - TCP Connection verify

Step 4: Authentication

Authentication was added using token OPS-0970. Controller sends AUTH command first, & agent whether the token is correct. A valid token returns OK AUTHENTICATED SID: 0790, while invalid token returns an authentication error.

```

mindio4@localhost:~$
mindio4@localhost:~$ cd RemoteOps
mindio4@localhost:~/RemoteOps$ nano agent_970.c
mindio4@localhost:~/RemoteOps$ nano controller_970.c

mindio4@localhost:~/RemoteOps$ nano agent_970.c
mindio4@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindio4@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:51312
Received command: AUTH OPS-0970
Authentication successful.
Connection closed.
mindio4@localhost:~/RemoteOps$ |

```

Figure 7- Received AUTH Clarification

```
mindy04@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970
mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Sent: AUTH OPS-0970
Agent response: OK AUTHENTICATED SID:0790
mindy04@localhost:~/RemoteOps$ |
```

Figure 8 - Agent Response

Controller code AUTH-TOKEN

The **AUTH-TOKEN** value in the controller code was modified as required. After updating token, controller code was executed to verify the changes.

```
GNU nano 8.1 controller_970.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define SERVER_IP "127.0.0.1"
#define PORT 9410
#define AUTH_TOKEN "OPS-0000"

int main(void)
{
    int sockfd;
    struct sockaddr_in server_addr;

    char buffer[1024];

    sockfd = socket(AF_INET, SOCK_STREAM, 0);

    if (sockfd < 0)
    {
        perror("socket");
        return 1;
    }

    memset(&server_addr, 0, sizeof(server_addr));

    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);

    if (inet_pton(AF_INET,
                  SERVER_IP,
                  &server_addr.sin_addr) <= 0)
    {
        perror("inet_pton");
        close(sockfd);
    }
}
```

Figure 9 - change AUTH_TOKEN - OPS-0000

```

mindio4@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:46342
Received command: AUTH OPS-0000
Authentication failed.
Connection closed.
mindio4@localhost:~/RemoteOps$ |

```

Figure 10 - Server received failed text

```

mindio4@localhost:~/RemoteOps$ nano controller_970.c
mindio4@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970
mindio4@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Sent: AUTH OPS-0000
Agent response: ERR 001 AUTH_FAILED SID:0790
mindio4@localhost:~/RemoteOps$ |

```

Figure 11 - client send modify AUTH_TOKEN

Step 5 — Persistent Command Loop + SYSINFO + QUIT

Allowing controller to send multiple commands through the same connection without reconnecting after each command.

SYSINFO command implemented to retrieve system information like CPU load, memory usage & system uptime.

Agent reads values from filesystem & sends collected info to controller along with unique session ID (SID).

```

mindio4@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindio4@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:39268
Received: AUTH OPS-0970
Authentication successful.
Received command: SYSINFO
Received command: SYSINFO
Received command: SYSINFO
Received command: HELLO
Received command: QUIT
Controller requested QUIT.
Connection closed.
mindio4@localhost:~/RemoteOps$ |

```

Figure 12 - server received command

```

mindio4@localhost:~/RemoteOps$ nano controller_970.c
mindio4@localhost:~/RemoteOps$ nano controller_970.c
mindio4@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970
mindio4@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> SYSINFO
Agent: OK SYSINFO 0.04 1449 6377 SID:0790

remoteops> SYSINFO
Agent: OK SYSINFO 0.01 1423 6436 SID:0790

remoteops> SYSINFO
Agent: OK SYSINFO 0.01 1423 6448 SID:0790

remoteops> HELLO
Agent: ERR 003 UNKNOWN_COMMAND SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindio4@localhost:~/RemoteOps$ |

```

Figure 13 - client connected to server via SID

Step 6 — LISTPROC

This command implemented to retrieve & display snapshot of running processes on system.

Agent uses popen() function to execute Linux ps command & read the process information.

Returns process ID & names to controller in the required format.

```

mindio4@localhost:~/RemoteOps$ nano agent_970.c
mindio4@localhost:~/RemoteOps$ ps -eo pid,comm --no-headers
 1 systemd
 2 kthreadd
 3 pool_workqueue_release
 4 kworker/R-rcu_gp
 5 kworker/R-sync_wq
 6 kworker/R-kvfree_rcu_reclaim
 7 kworker/R-slub_flushwq
 8 kworker/R-netns
12 kworker/u8:0-events_unbound
13 kworker/u8:1-ipv6_addrconf
14 kworker/R-mm_percpu_wq
15 ksoftirqd/0
16 rcu_preempt
17 rcu_exp_par_gp_kthread_worker/0
18 rcu_exp_gp_kthread_worker
19 migration/0
20 idle_inject/0
21 cpuhp/0
22 cpuhp/1
23 idle_inject/1
24 migration/1
25 ksoftirqd/1

```

Figure 14 - Verify of running process using Linux ps command

```

mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> LISTPROC
Agent: OK PROCS      1 systemd,      2 kthreadd,      3 pool_workqueue_release,      4 kworker/R-rcu_
gp,      5 kworker/R-sync_wq,      6 kworker/R-kvfree_rcu_reclaim,      7 kworker/R-slub_flushwq,
8 kworker/R-netns,      12 kworker/u8:0-events_unbound,      13 kworker/u8:1-ipv6_addrconf,      14 kworker/R-mm_percpu_wq,
15 ksoftirqd/0,      16 rcu_preempt,      17 rcu_exp_par_gp_kthread_worker/0,
18 rcu_exp_gp_kthread_worker,      19 migration/0,      20 idle_inject/0,      21 cpuhp/0,      22 cpuhp
/1,      23 idle_inject/1,      24 migration/1,      25 ksoftirqd/1,      27 kworker/1:0H-kblockd,      32
kdevtmpfs,      33 kworker/R-inet_frag_wq,      34 rcu_tasks_kthread,      35 rcu_tasks_rude_kthread,
36 rcu_tasks_trace_kthread,      37 kauditd,      38 khungtaskd,      39 oom_reaper,      41 kworker/R-wr
iteback,      42 kcompactd0,      43 ksm,      44 kh SID:0790

remoteops> SYSINFO
Agent: OK SYSINFO 0.14 1428 8593 SID:0790

remoteops> LISTPROC
Agent: OK PROCS      1 systemd,      2 kthreadd,      3 pool_workqueue_release,      4 kworker/R-rcu_
gp,      5 kworker/R-sync_wq,      6 kworker/R-kvfree_rcu_reclaim,      7 kworker/R-slub_flushwq,
8 kworker/R-netns,      12 kworker/u8:0-events_unbound,      13 kworker/u8:1-ipv6_addrconf,      14 kworker/R-mm_percpu_wq,
15 ksoftirqd/0,      16 rcu_preempt,      17 rcu_exp_par_gp_kthread_worker/0,
18 rcu_exp_gp_kthread_worker,      19 migration/0,      20 idle_inject/0,      21 cpuhp/0,      22 cpuhp
/1,      23 idle_inject/1,      24 migration/1,      25 ksoftirqd/1,      27 kworker/1:0H-kblockd,      32
kdevtmpfs,      33 kworker/R-inet_frag_wq,      34 rcu_tasks_kthread,      35 rcu_tasks_rude_kthread,
36 rcu_tasks_trace_kthread,      37 kauditd,      38 khungtaskd,      39 oom_reaper,      41 kworker/R-wr
iteback,      42 kcompactd0,      43 ksm,      44 kh SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 15 - Testing of LISTPROC & SYSINFO commands

```

mindy04@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindy04@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:57580
Received: AUTH OPS-0790
Authentication successful.
Received command: LISTPROC
Received command: SYSINFO
Received command: LISTPROC
Received command: QUIT
Controller requested QUIT.
Connection closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 16 - Successful LISTPROC & SYSINFO execution

Step 7 — Restricted EXEC Command.

EXEC command implemented using DATE, UPTIME, DISKFREE, HOSTNAME & WHOAMI fixed whitelist. Agent allows execution only for commands included in predefined whitelist.

Any command like EXEC LS, is rejected to prevent unrestricted remote command execution.

```

mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> EXEC DATE
Agent: OK EXEC_RESULT Tue Oct 6 01:17:25 AM +0530 2026 SID:0790

remoteops> EXEC HOSTNAME
Agent: OK EXEC_RESULT localhost.localdomain SID:0790

remoteops> EXEC WHOAMI
Agent: OK EXEC_RESULT mindy04 SID:0790

remoteops> EXEC LS
Agent: ERR 002 COMMAND_NOT_ALLOWED SID:0790

remoteops> EXEC RM
Agent: ERR 002 COMMAND_NOT_ALLOWED SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 17 - Testing restricted EXEC commands

Successful testing restricted EXEC features & returns the expected results with SID: 0790. An unauthorized command EXEC LS was rejected with ERR 002 COMMAND_NOT_ALLOWED confirming restriction works correctly.

```

mindy04@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindy04@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:39676
Received: AUTH OPS-0970
Authentication successful.
Received command: EXEC DATE
Received command: EXEC HOSTNAME
Received command: EXEC WHOAMI
Received command: EXEC LS
Received command: EXEC RM
Received command: QUIT
Controller requested QUIT.
Connection closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 18 - Agent-Side Execution and Rejection of Restricted EXEC Commands

Agent receiving and processing several EXEC commands after successful authentication. While unauthorized commands such as EXEC LS and EXEC RM were rejected. Agent then processed the QUIT command and closed the Controller connection successfully.

Step 8 – PUT file upload

PUT was implemented to allow controller to upload a local file to agent. Controller determines file size & sends filename & filesize to agent before transmitting raw file data.

Writes data to file, & stores uploaded file in designated directory, ./agentfiles/IT24100970/

Agent side, file is transferred accurately and stored correctly.

```
mindio4@localhost:~/RemoteOps$ nano agent_970

mindio4@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindio4@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:54026
Received: AUTH OPS-0970
Authentication successful.
Received command: PUT test.txt 42
Receiving file: test.txt (42 bytes)
File received successfully: ./agentfiles/IT24100970/test.txt
Received command: QUIT
Controller requested QUIT.
Connection closed.
mindio4@localhost:~/RemoteOps$ sha256sum test.txt agentfiles/IT24100970/test.txt
0e00fa0fd3793362ee78625232cd2e5c4cf4626da73da7650c38175330964519  test.txt
0e00fa0fd3793362ee78625232cd2e5c4cf4626da73da7650c38175330964519  agentfiles/IT2410
0970/test.txt
mindio4@localhost:~/RemoteOps$ |
```

Figure 19 - Successful PUT File Upload and SHA-256 Verification

Agent received the 42-byte file and stored it in the personalized directory ./agentfiles/IT24100970/test.txt. SHA-256 hashes for the original and uploaded files confirmed that the file content remained unchanged during transfer.

```
mindio4@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970
mindio4@localhost:~/RemoteOps$ echo "RemoteOps file upload test for IT24100970" > test.txt
mindio4@localhost:~/RemoteOps$ cat test.txt
RemoteOps file upload test for IT24100970
mindio4@localhost:~/RemoteOps$ wc -c test.txt
42 test.txt
mindio4@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> PUT test.txt
Uploading test.txt (42 bytes)...
Agent: OK FILE_RECEIVED test.txt SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindio4@localhost:~/RemoteOps$ ls -l agentfiles/IT24100970/
total 4
-rw-r--r--. 1 mindio4 mindio4 42 Oct  6 23:01 test.txt
mindio4@localhost:~/RemoteOps$ cat agentfiles/IT24100970/test.txt
RemoteOps file upload test for IT24100970
mindio4@localhost:~/RemoteOps$ cmp test.txt agentfiles/IT24100970/test.txt
mindio4@localhost:~/RemoteOps$ |
```

Figure 20 - Successful PUT File Upload from Controller to Agent

Step 9 – GET upload

GET was implemented to allow the Controller to retrieve files previously stored on the Agent. When a file is requested, the Agent sends the **filename and file size**, followed by the raw file data to the Controller.

```
mindio4@localhost:~/RemoteOps$ nano agent_970.c
mindio4@localhost:~/RemoteOps$ nano controller_970.c
mindio4@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindio4@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970
mindio4@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> GET test.txt
Downloading test.txt (42 bytes)...
File saved as: downloaded_test.txt

remoteops> GET abcd.txt
Agent: ERR 005 FILE_NOT_FOUND SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindio4@localhost:~/RemoteOps$ |
```

Figure 21- Successful GET File Upload from Controller to Agent

```
mindio4@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970
mindio4@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:37386
Received: AUTH OPS-0970
Authentication successful.
Received command: GET test.txt
Sending file: test.txt (42 bytes)
File sent successfully: ./agentfiles/IT24100970/test.txt
Received command: GET abcd.txt
Received command: QUIT
Controller requested QUIT.
Connection closed.
mindio4@localhost:~/RemoteOps$ ls -l agentfiles/IT24100970/
total 4
-rw-r--r--. 1 mindio4 mindio4 42 Oct  6 23:01 test.txt
mindio4@localhost:~/RemoteOps$ cat downloaded_test.txt
RemoteOps file upload test for IT24100970
mindio4@localhost:~/RemoteOps$ cat test.txt
RemoteOps file upload test for IT24100970
mindio4@localhost:~/RemoteOps$ cmp test.txt downloaded_test.txt
mindio4@localhost:~/RemoteOps$ |
```

Figure 22 - Successful GET File Download and File Integrity Verification

Agent successfully processing the GET test.txt request and sending the 42-byte file from the personalized storage directory. A request for a non-existing file, GET abcd.txt, was also tested to verify error handling.

Step 10 – UDP Monitoring

UDP monitoring functionality was implemented to allow the Agent to periodically send monitoring information to the Controller using the UDP protocol.

```
Controller closed.
mindie4@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970 -pthread
mindie4@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> MONITOR START
Agent: OK MONITOR_STARTED SID:0790

remoteops>
[UDP Monitor] SYSINFO 0.15 2540 92978 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.14 2540 92981 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.14 2540 92984 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.13 2540 92987 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.12 2540 92990 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.12 2540 92993 SID:0790
remoteops> MON
[UDP Monitor] SYSINFO 0.11 2540 92996 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.11 2540 92999 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.18 2540 93002 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.16 2540 93005 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.16 2540 93008 SID:0790
remoteops> MONITOR STOP
Agent: ERR 003 UNKNOWN_COMMAND SID:0790

remoteops>
[UDP Monitor] SYSINFO 0.15 2540 93011 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.15 2540 93014 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.14 2540 93017 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.13 2540 93020 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.13 2540 93023 SID:0790
remoteops>
[UDP Monitor] SYSINFO 0.12 2540 93026 SID:0790
remoteops> QUIT
[UDP Monitor] SYSINFO 0.12 2540 93029 SID:0790
remoteops>
Agent: OK BYE SID:0790
Controller closed.
mindie4@localhost:~/RemoteOps$ |
```

Figure 23 - UDP Periodic System Monitor

Confirms that monitoring data was received continuously through the UDP channel while the TCP session remained active.

```

mindy04@localhost:~/RemoteOps$ gcc agent_970.c -o agent_970 -pthread
mindy04@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Controller connected from 127.0.0.1:34148
Received: AUTH OPS-0970
Authentication successful.
Received command: MONITOR START 9500
UDP monitoring started on port 9500.
UDP monitoring thread started for 127.0.0.1:9500
Received command: MONMONITOR STOP
Received command: QUIT
UDP monitoring thread stopped.
Controller requested QUIT.
Connection closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 24 -Successful MONITOR START and UDP SYSINFO

MONITOR STOP command in this test was not recognized and required further correction.

Step 11 – Multi –Client Concurrency

The Agent was modified to support multiple Controller connections at the same time using POSIX threads. Each Controller is handled by a separate thread, allowing several users to authenticate and execute commands independently without blocking one another.

```

mindy04@localhost:~/RemoteOps$ ./agent_970
RemoteOps Agent listening on port 9410...
Multi-client mode enabled.
Controller connected from 127.0.0.1:51076
Received: AUTH OPS-0970
Authentication successful.
Received command: SYSINFO
Controller connected from 127.0.0.1:39392
Received: AUTH OPS-0970
Authentication successful.
Received command: LISTPROC
Controller connected from 127.0.0.1:42112
Received: AUTH OPS-0970
Authentication successful.
Received command: EXEC DATE
Controller connected from 127.0.0.1:42618
Received: AUTH OPS-0970
Authentication successful.
Received command: EXEC WHOAMI
Received command: trSYSINFO
Received command: QUIT
Controller requested QUIT.
Controller session closed: 127.0.0.1:42618
^C
mindy04@localhost:~/RemoteOps$ |

```

Figure 25 - Multi-Client Concurrency Testing Using POSIX Threads

Each client authenticated and executed different commands independently, confirming that the thread-per-client concurrency model was working correctly.

```

mindy04@localhost:~$ cd RemoteOps
mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> LISTPROC
Agent: OK PROCS      1 systemd,      2 kthreadd,      3 pool_workqueue_rele
ase,      4 kworker/R-rcu_gp,      5 kworker/R-sync_wq,      6 kworker/R-kvf
ree_rcu_reclaim,      7 kworker/R-slub_flushwq,      8 kworker/R-netns,
12 kworker/u8:0-events_unbound,      13 kworker/u8:1-ipv6_addrconf,      14 kw
orker/R-mm_percpu_wq,      15 ksoftirqd/0,      16 rcu_preempt,      17 rcu_exp
_par_gp_kthread_worker/0,      18 rcu_exp_gp_kthread_worker,      19 migration
/0,      20 idle_inject/0,      21 cpuhp/0,      22 cpuhp/1,      23 idle_inject
/1,      24 migration/1,      25 ksoftirqd/1,      32 kdevtmpfs,      33 kworker
/R-inet_frag_wq,      34 rcu_tasks_kthread,      35 rcu_tasks_rude_kthread,
      36 rcu_tasks_trace_kthread,      37 kauditd,      38 khungtaskd,      39 oom_
reaper,      41 kworker/R-writeback,      42 kcompactd0,      43 ksm,      44 k
hugepaged,      45 kworker/R-cr SID:0790

remoteops> |

```

Figure 26 - Controller 1 execution

```

Controller closed.
mindy04@localhost:~/RemoteOps$ gcc controller_970.c -o controller_970 -pthread
mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> SYSINFO
Agent: OK SYSINFO 0.19 2536 94228 SID:0790

remoteops> LISTPROC
Agent: OK PROCS      1 systemd,      2 kthreadd,      3 pool_workqueue_release,      4 kworker/R-rcu_
gp,      5 kworker/R-sync_wq,      6 kworker/R-kvfree_rcu_reclaim,      7 kworker/R-slub_flushwq,
      8 kworker/R-netns,      12 kworker/u8:0-events_unbound,      13 kworker/u8:1-ipv6_addrconf,      14 kwor
ker/R-mm_percpu_wq,      15 ksoftirqd/0,      16 rcu_preempt,      17 rcu_exp_par_gp_kthread_worker/0,
      18 rcu_exp_gp_kthread_worker,      19 migration/0,      20 idle_inject/0,      21 cpuhp/0,      22 cpuhp
/1,      23 idle_inject/1,      24 migration/1,      25 ksoftirqd/1,      32 kdevtmpfs,      33 kworker/R-i
net_frag_wq,      34 rcu_tasks_kthread,      35 rcu_tasks_rude_kthread,      36 rcu_tasks_trace_kthread,
      37 kauditd,      38 khungtaskd,      39 oom_reaper,      41 kworker/R-writeback,      42 kcompactd0,
      43 ksm,      44 khugepaged,      45 kworker/R-cr SID:0790

remoteops> EXEC DATE
Agent: OK EXEC_RESULT Wed Oct  7 12:47:45 AM +0530 2026 SID:0790

remoteops> EXEC WHOAMI
Agent: OK EXEC_RESULT mindy04 SID:0790

remoteops> SYSINFO
Agent: OK SYSINFO 0.11 2536 94300 SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.

```

Figure 27- Controller 2 execution

```

Controller closed.
mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> EXEC DATE
Agent: OK EXEC_RESULT Wed Oct  7 01:15:58 AM +0530 2026 SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 28 - Controller 3 execution

```

mindy04@localhost:~$
mindy04@localhost:~$ cd RemoteOps
mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> EXEC WHOAMI
Agent: OK EXEC_RESULT mindy04 SID:0790

remoteops> SYSINFO
Agent: ERR 003 UNKNOWN_COMMAND SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 29 - Controller 4 execution

```

mindy04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> GET test.txt
Downloading test.txt (42 bytes)...
File saved as: downloaded_test.txt

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindy04@localhost:~/RemoteOps$ |

```

Figure 30 - Controller 5 execution

Step 12 - Timestamped Logging

A logging system was added to record important activities in `remoteops_IT24100970.log`. The log records connections, authentication results, commands, file transfers, and disconnects with timestamps. A mutex is used to protect the log file when multiple client threads write to it at the same time.

```

mindio04@localhost:~/RemoteOps$ cat remoteops_IT24100970.log
2026-10-07 01:15:04 AGENT STARTED
2026-10-07 01:15:06 CONNECT 127.0.0.1:46904
2026-10-07 01:15:06 AUTH SUCCESS 127.0.0.1:46904
2026-10-07 01:15:22 COMMAND 127.0.0.1:46904 SYSINFO
2026-10-07 01:15:26 COMMAND 127.0.0.1:46904 QUIT
2026-10-07 01:15:26 DISCONNECT 127.0.0.1:46904
2026-10-07 01:15:29 CONNECT 127.0.0.1:45210
2026-10-07 01:15:29 AUTH SUCCESS 127.0.0.1:45210
2026-10-07 01:15:36 COMMAND 127.0.0.1:45210 LISTPROC
2026-10-07 01:15:40 COMMAND 127.0.0.1:45210 QUIT
2026-10-07 01:15:40 DISCONNECT 127.0.0.1:45210
2026-10-07 01:15:45 CONNECT 127.0.0.1:36056
2026-10-07 01:15:45 AUTH SUCCESS 127.0.0.1:36056
2026-10-07 01:15:58 COMMAND 127.0.0.1:36056 EXEC DATE
2026-10-07 01:16:01 COMMAND 127.0.0.1:36056 QUIT
2026-10-07 01:16:01 DISCONNECT 127.0.0.1:36056
2026-10-07 01:16:06 CONNECT 127.0.0.1:58638
2026-10-07 01:16:06 AUTH SUCCESS 127.0.0.1:58638
2026-10-07 01:16:13 COMMAND 127.0.0.1:58638 PUT test.txt 42
2026-10-07 01:16:13 PUT test.txt 42 bytes from 127.0.0.1:58638
2026-10-07 01:16:16 COMMAND 127.0.0.1:58638 QUIT
2026-10-07 01:16:16 DISCONNECT 127.0.0.1:58638
2026-10-07 01:16:37 CONNECT 127.0.0.1:51050
2026-10-07 01:16:37 AUTH SUCCESS 127.0.0.1:51050
2026-10-07 01:16:40 COMMAND 127.0.0.1:51050 GET test.txt
2026-10-07 01:16:40 GET test.txt 42 bytes to 127.0.0.1:51050
2026-10-07 01:16:43 COMMAND 127.0.0.1:51050 QUIT
2026-10-07 01:16:43 DISCONNECT 127.0.0.1:51050
mindio04@localhost:~/RemoteOps$

```

Figure 31 - Timestamped RemoteOps Activity Log

The personalized log file records important RemoteOps events with timestamps, including connections, authentication, commands, file transfers, and disconnections. This confirms that the Agent logging function is working correctly.

Step 13 - Makefile and Final Testing

The personalized Makefile_970 was created to compile both the Agent and Controller using GCC with pthread support. Final regression testing was carried out for authentication, SYSINFO, LISTPROC, EXEC, PUT, GET, UDP monitoring, concurrency, logging, and QUIT to confirm that all mandatory features worked together correctly.

```

mindy04@localhost:~/RemoteOps$
mindy04@localhost:~/RemoteOps$ nano Makefile_970
mindy04@localhost:~/RemoteOps$ make -f Makefile_970 clean
rm -f agent_970 controller_970
mindy04@localhost:~/RemoteOps$ make -f Makefile_970
gcc -Wall -Wextra -pthread agent_970.c -o agent_970
agent_970.c: In function 'handle_client':
agent_970.c:913:28: warning: '%s' directive output may be truncated writing up to 1
023 bytes into a region of size between 482 and 501 [-Wformat-truncation=]
  913 |         "COMMAND %s:%d %s",
      |                        ^~
.....
  916 |         buffer);
      |         ~~~~~
agent_970.c:910:9: note: 'snprintf' output between 12 and 1054 bytes into a destina
tion of size 512
  910 |         snprintf(
      |         ~~~~~
  911 |             log_message,
      |             ~~~~~
  912 |             sizeof(log_message),
      |             ~~~~~
  913 |             "COMMAND %s:%d %s",
      |             ~~~~~
  914 |             monitor.client_ip,
      |             ~~~~~
  915 |             client_port,
      |             ~~~~~
  916 |             buffer);
      |             ~~~~~

```

Figure 32 - Building RemoteOps Using the Unique Makefile

```

agent_970.c:910:9: note: 'snprintf' output between 12 and 1054 bytes into a destina
tion of size 512
  910 |         snprintf(
      |         ~~~~~
  911 |             log_message,
      |             ~~~~~
  912 |             sizeof(log_message),
      |             ~~~~~
  913 |             "COMMAND %s:%d %s",
      |             ~~~~~
  914 |             monitor.client_ip,
      |             ~~~~~
  915 |             client_port,
      |             ~~~~~
  916 |             buffer);
      |             ~~~~~

gcc -Wall -Wextra -pthread controller_970.c -o controller_970
controller_970.c: In function 'main':
controller_970.c:1059:18: warning: 'snprintf' output may be truncated before the la
st format character [-Wformat-truncation=]
 1059 |         "%s\n",
      |         ^
controller_970.c:1056:9: note: 'snprintf' output between 2 and 1025 bytes into a de
stination of size 1024
 1056 |         snprintf(
      |         ~~~~~
 1057 |             protocol_command,
      |             ~~~~~
 1058 |             sizeof(protocol_command),
      |             ~~~~~
 1059 |             "%s\n",
      |             ~~~~~
 1060 |             command);
      |             ~~~~~

```

Figure 33 - Compiler Warning Details

```

mindi04@localhost:~/RemoteOps$ ss -tlnp | grep 9410
LISTEN 0      10          0.0.0.0:9410  0.0.0.0:*    users:(("agent_970",pid=39241,fd=3))
mindi04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> SYSINFO
Agent: OK SYSINFO 0.00 2576 128046 SID:0790

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindi04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> GET missing.txt
Agent: ERR 005 FILE_NOT_FOUND SID:0790

remoteops> GET test.txt
Downloading test.txt (42 bytes)...
File saved as: downloaded_test.txt

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindi04@localhost:~/RemoteOps$ ./controller_970
Connected to RemoteOps Agent.
Agent: OK AUTHENTICATED SID:0790

remoteops> GET test.txt
Downloading test.txt (42 bytes)...
File saved as: downloaded_test.txt

remoteops> QUIT
Agent: OK BYE SID:0790
Controller closed.
mindi04@localhost:~/RemoteOps$ |

```

Figure 34 - Final Functional Testing and Verification of TCP Port 9410

```

mindi04@localhost:~/RemoteOps$ cmp test.txt downloaded_test.txt
mindi04@localhost:~/RemoteOps$ cat remoteops_IT24100970.log
2026-10-07 01:15:04 AGENT STARTED
2026-10-07 01:15:06 CONNECT 127.0.0.1:46904
2026-10-07 01:15:06 AUTH SUCCESS 127.0.0.1:46904
2026-10-07 01:15:22 COMMAND 127.0.0.1:46904 SYSINFO
2026-10-07 01:15:26 COMMAND 127.0.0.1:46904 QUIT
2026-10-07 01:15:26 DISCONNECT 127.0.0.1:46904
2026-10-07 01:15:29 CONNECT 127.0.0.1:45210
2026-10-07 01:15:29 AUTH SUCCESS 127.0.0.1:45210
2026-10-07 01:15:36 COMMAND 127.0.0.1:45210 LISTPROC
2026-10-07 01:15:40 COMMAND 127.0.0.1:45210 QUIT
2026-10-07 01:15:40 DISCONNECT 127.0.0.1:45210
2026-10-07 01:15:45 CONNECT 127.0.0.1:36056
2026-10-07 01:15:45 AUTH SUCCESS 127.0.0.1:36056
2026-10-07 01:15:58 COMMAND 127.0.0.1:36056 EXEC DATE
2026-10-07 01:16:01 COMMAND 127.0.0.1:36056 QUIT
2026-10-07 01:16:01 DISCONNECT 127.0.0.1:36056
2026-10-07 01:16:06 CONNECT 127.0.0.1:58638
2026-10-07 01:16:06 AUTH SUCCESS 127.0.0.1:58638
2026-10-07 01:16:13 COMMAND 127.0.0.1:58638 PUT test.txt 42
2026-10-07 01:16:13 PUT test.txt 42 bytes from 127.0.0.1:58638
2026-10-07 01:16:16 COMMAND 127.0.0.1:58638 QUIT
2026-10-07 01:16:16 DISCONNECT 127.0.0.1:58638
2026-10-07 01:16:37 CONNECT 127.0.0.1:51050
2026-10-07 01:16:37 AUTH SUCCESS 127.0.0.1:51050
2026-10-07 01:16:40 COMMAND 127.0.0.1:51050 GET test.txt
2026-10-07 01:16:40 GET test.txt 42 bytes to 127.0.0.1:51050
2026-10-07 01:16:43 COMMAND 127.0.0.1:51050 QUIT
2026-10-07 01:16:43 DISCONNECT 127.0.0.1:51050
2026-10-07 10:09:59 AGENT STARTED
2026-10-07 10:10:26 CONNECT 127.0.0.1:51168
2026-10-07 10:10:27 AUTH SUCCESS 127.0.0.1:51168
2026-10-07 10:10:35 COMMAND 127.0.0.1:51168 SYSINFO
2026-10-07 10:10:39 COMMAND 127.0.0.1:51168 QUIT
2026-10-07 10:10:39 DISCONNECT 127.0.0.1:51168
2026-10-07 10:10:43 CONNECT 127.0.0.1:33344
2026-10-07 10:10:43 AUTH SUCCESS 127.0.0.1:33344
2026-10-07 10:10:51 COMMAND 127.0.0.1:33344 LISTPROC
2026-10-07 10:10:55 COMMAND 127.0.0.1:33344 QUIT
2026-10-07 10:10:55 DISCONNECT 127.0.0.1:33344
2026-10-07 10:11:06 CONNECT 127.0.0.1:42508
2026-10-07 10:11:06 AUTH SUCCESS 127.0.0.1:42508
2026-10-07 10:11:08 COMMAND 127.0.0.1:42508 EXEC DATE
2026-10-07 10:11:12 COMMAND 127.0.0.1:42508 QUIT
2026-10-07 10:11:12 DISCONNECT 127.0.0.1:42508
2026-10-07 10:11:23 CONNECT 127.0.0.1:43940
2026-10-07 10:11:23 AUTH SUCCESS 127.0.0.1:43940
2026-10-07 10:11:25 COMMAND 127.0.0.1:43940 EXEC HOSTNAME
2026-10-07 10:11:28 COMMAND 127.0.0.1:43940 QUIT
2026-10-07 10:11:28 DISCONNECT 127.0.0.1:43940
2026-10-07 10:12:17 CONNECT 127.0.0.1:39854
2026-10-07 10:12:17 AUTH SUCCESS 127.0.0.1:39854
2026-10-07 10:12:28 COMMAND 127.0.0.1:39854 GET missing.txt
2026-10-07 10:12:45 COMMAND 127.0.0.1:39854 GET test.txt
2026-10-07 10:12:45 GET test.txt 42 bytes to 127.0.0.1:39854
2026-10-07 10:12:48 COMMAND 127.0.0.1:39854 QUIT
2026-10-07 10:12:48 DISCONNECT 127.0.0.1:39854
2026-10-07 10:13:08 CONNECT 127.0.0.1:58154
2026-10-07 10:13:08 AUTH SUCCESS 127.0.0.1:58154
2026-10-07 10:13:15 COMMAND 127.0.0.1:58154 GET test.txt
2026-10-07 10:13:15 GET test.txt 42 bytes to 127.0.0.1:58154
2026-10-07 10:13:22 COMMAND 127.0.0.1:58154 QUIT
2026-10-07 10:13:22 DISCONNECT 127.0.0.1:58154
2026-10-07 10:13:34 CONNECT 127.0.0.1:44282
2026-10-07 10:13:34 AUTH SUCCESS 127.0.0.1:44282
2026-10-07 10:13:36 COMMAND 127.0.0.1:44282 MONITOR START 9500
2026-10-07 10:13:44 COMMAND 127.0.0.1:44282 MONITOR STOP
2026-10-07 10:13:48 COMMAND 127.0.0.1:44282 QUIT
2026-10-07 10:13:48 DISCONNECT 127.0.0.1:44282
mindi04@localhost:~/RemoteOps$ |

```

Figure 35, 36 - File Integrity and Activity Log Verification

Testing Summary

Test	Input	Expected Result	Actual Result	Status
Authentication	AUTH OPS-0970	OK AUTHENTICATED SID:0790	Authentication accepted	Pass
Invalid authentication	Incorrect token	ERR 001 AUTH_FAILED SID:0790	Invalid token rejected	Pass
SYSINFO	SYSINFO	CPU, memory, uptime + SID	Values returned	Pass
LISTPROC	LISTPROC	Process snapshot + SID	Process list returned	Pass
Allowed EXEC	EXEC DATE / WHOAMI etc.	OK EXEC_RESULT	All five allowed commands worked	Pass
Blocked EXEC	EXEC LS	ERR 002 COMMAND_NOT_ALLOWED	Rejected correctly	Pass
PUT	PUT test.txt	File stored in personalized directory	Upload succeeded	Pass
GET	GET test.txt	Downloaded file identical	cmp/SHA-256 matched	Pass
UDP monitoring	MONITOR START / STOP	Periodic UDP SYSINFO then stop	Datagrams received and stopped	Pass
Concurrency	5 Controllers simultaneously	All sessions remain responsive	Five sessions handled	Pass
Logging	Run commands and inspect log	Timestamped events written	Log entries created	Pass
QUIT	QUIT	OK BYE SID:0790 and clean close	Session closed cleanly	Pass

Design Rationale and Assumptions

- **Thread-per-client concurrency:** Chosen because it provides a simple mapping between one Controller session and one execution context while allowing multiple clients to work independently.
- **TCP for control and file transfer:** Used because authentication, commands and files require reliable, ordered delivery.
- **UDP for monitoring:** Used because periodic monitoring is lightweight and the assignment explicitly requires a secondary UDP channel.
- **Line-based framing:** `recv_line()` follows the protocol newline delimiter and avoids assuming that one `recv()` equals one complete command.
- **Exact file byte counting:** PUT and GET loop until the declared filesize has been transferred because TCP can split data across multiple `send ()/recv()` calls.
- **Fixed EXEC whitelist:** Prevents arbitrary remote shell execution and matches the assignment safety requirement.
- **Mutex-protected logging:** Required because multiple client threads can write to the same log file concurrently.

Development Process Evidence

The implementation was developed incrementally and pushed to GitHub after testing each major feature. The commit history contains separate commits for project setup, TCP Agent, TCP Controller, authentication, SYSINFO, LISTPROC, restricted EXEC, PUT, GET, UDP monitoring, concurrency and logging. This provides a clear development trail rather than a single last-minute commit.

Conclusion

RemoteOps was implemented as a C-based client/server monitoring and management tool using the BSD sockets API. The final system supports authenticated TCP control, system information, process listing, restricted remote execution, exact-byte file upload/download, periodic UDP monitoring, graceful disconnect handling, multi-client concurrency and timestamped logging. Personalized values derived from IT24100970 are used throughout the implementation. Final regression testing confirmed that the mandatory features work together correctly.

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For example,

Figure 2.6: XYZ Connector

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